

One CSI Report Cannot Serve Every Consumer: The Consumer-Relative False-Clear Rate in 5G/6G Link Adaptation

Andrew H. Bond

Department of Computer Engineering
San José State University, San José, CA, USA
andrew.bond@sjsu.edu

Abstract—A channel-state-information (CSI) report is a certificate: an implicit claim that the channel is good enough to transmit at a selected rate. That certificate is not read by the channel; it is read by a *consumer*—a scheduler with a reliability target and a decoder at a moment in time. We make the rate at which a CSI certificate *false-clears* (selects a rate the link cannot support) a first-class, measured KPI on a real 5G NR link-level chain (Sionna LDPC decoding), and show it is *consumer-relative* along two axes. First, the reliability target: the identical CSI-derived modulation-and-coding selection that meets an eMBB target (BLER = 10^{-1}) misses a URLLC target (10^{-3}) by more than two orders of magnitude ($112\times$), while a target-aware policy meets it. Second, the refresh horizon: as the report ages, the false-clear rate grows, and the largest safe report period is proportional to the Clarke coherence time ($\approx 0.177 T_{\text{coh}}$, $R^2 = 0.915$). The witnessed corrections that hold each axis—outer-loop link adaptation, a coherence-driven refresh—already exist; our contribution is the measured number that says when each is needed. All results are drawn from sealed, pre-registered records.

Index Terms—CSI feedback, link adaptation, URLLC, outer-loop link adaptation, reliability, 5G NR.

I. INTRODUCTION

Adaptive modulation and coding (AMC) turns a channel-state-information (CSI) report into a rate decision: the transmitter selects the highest modulation-and-coding scheme (MCS) whose required SINR the reported CSI clears. The report is thus a *certificate*—an implicit assertion that the link will decode at the chosen rate. Like any certificate, it can be wrong in a way that matters only to whoever reads it. We call a wrong-in-the-optimistic-direction certificate a *false-clear*: the CSI cleared an MCS that the true channel cannot support, and the transport fails (a HARQ NACK, a block error).

Our thesis is that the false-clear rate is not a property of the channel or even of the report alone; it is a property of the report *and its consumer*. Two schedulers reading the same CSI can disagree on whether it false-clears, because they read it against different requirements. We make the false-clear rate a measured KPI on a real 5G NR link-level chain and demonstrate this consumer-relativity along two operationally important axes:

- **the reliability target (§III)**: a single CSI-derived MCS that is adequate for enhanced mobile broadband (eMBB)

is catastrophically optimistic for ultra-reliable low-latency communication (URLLC), by more than $100\times$;

- **the refresh horizon (§IV)**: the false-clear rate grows as the certificate ages, and the safe refresh period scales with the channel coherence time by a simple measured law.

Neither observation requires a new estimator or scheduler. The corrections are known: outer-loop link adaptation (OLLA) [1] closes the target-relativity gap, and coherence-driven CSI reporting closes the refresh gap. What has been missing is the *measured, consumer-relative false-clear rate* that quantifies when each correction is load-bearing and how large the exposure is when it is absent.

II. METHOD

Substrate. We use a real 5G NR link-level chain in Sionna [2]: LDPC-coded QAM over the NR PDSCH, with block error rate (BLER) measured from the actual 5G LDPC decoder rather than an SINR-to-BLER formula. We first measure a real BLER(SNR) curve per MCS (400 coded blocks per SNR point over the grid -8 to 28 dB in 2 dB steps), then drive a per-TTI SNR trace through a 3GPP TR38.901 TDL-A channel at carrier 3.5 GHz with a chosen Doppler, and evaluate each policy's selected MCS against the true instantaneous SNR through the measured curve. The MCS set is the 3GPP NR table (TS 38.214 Table 5.1.3.1-1; QPSK through 64-QAM, code rates floored at $R \geq 1/5$ for the 5G LDPC base graph). The decoder is the *witness*: it independently determines whether the consumer decoded. Table I lists the parameters.

System model. Let $\gamma(t)$ (dB) be the true post-processing SNR at TTI t and $\hat{g}(t)$ the SNR the transmitter infers from the CSI report: for *fresh* CSI $\hat{g}(t) = \gamma(t) + n(t)$ with report noise $n \sim \mathcal{N}(0, \sigma^2)$; for a report *held* with period P , $\hat{g}(t) = \gamma(P\lfloor t/P \rfloor)$, so the estimate ages between reports. With MCS set $\mathcal{M} = \{(Q_m, R_m)\}$ and measured LDPC error curve $e_m(\gamma)$, the required SNR at target β is $\rho_m(\beta) = \min\{\gamma : e_m(\gamma) \leq \beta\}$. The AMC *certificate* selects the highest MCS clearing the estimate under a backoff Δ :

$$m^*(\hat{g}) = \max\{m \in \mathcal{M} : \rho_m(\beta_0) + \Delta \leq \hat{g}\}, \quad (1)$$

with reference target $\beta_0 = 10^{-1}$ and $\Delta = 0$ for the naive policy. The *consumer* is a reliability target β ; the *witness* is the LDPC decode of m^* at the *true* SNR. A false-clear is a cleared

TABLE I
LINK-LEVEL PARAMETERS (SEALED CELLS)

Parameter	Value
Carrier frequency	3.5 GHz
TTI duration	1 ms
Channel model	3GPP TR38.901 TDL-A
Mean SNR	12 dB
Decoder / code	Sionna 5G NR LDPC (real decode)
MCS set	NR TS 38.214 Tbl 5.1.3.1-1 (QPSK-64QAM)
BLER curves	400 blocks/pt, SNR -8-28 dB @ 2 dB
Trace length	6000 TTI
eMBB / URLLC target	10^{-1} / 10^{-3}
URLLC-aware policy	conservative MCS + $K=3$ diversity
Doppler (aging expt.)	200 Hz ($T_{\text{coh}} \approx 2.1$ ms)

MCS the channel cannot support, $\rho_{m^*}(\beta) > \gamma(t)$, and the measured, consumer-relative false-clear rate is the witnessed BLER of the selected MCS,

$$\text{FC}(\beta) = \frac{1}{T} \sum_{t=1}^T e_{m^*}(\hat{g}(t))(\gamma(t)), \quad (2)$$

read against β . Because (1)–(2) are judged against β , one trace $\{\hat{g}(t), \gamma(t)\}$ yields different verdicts for the eMBB ($\beta=10^{-1}$) and URLLC ($\beta=10^{-3}$) consumers—the consumer-relativity we measure. The URLLC-aware certificate raises Δ and combines K independent branches, failing only if all fail: $\text{FC}_{\text{aware}} = \frac{1}{T} \sum_t \prod_{k=1}^K e_{m^*}(\gamma_k(t))$. The witnessed *OLLA* correction (§IV) distrusts $\hat{g}$ and selects $m^*(\hat{g}(t) - o(t))$ with an offset driven by HARQ feedback,

$$o(t+1) = o(t) + \mu(1[\text{NACK}(t)] - \beta), \quad (3)$$

which drives the empirical error rate toward β under stale CSI. For held CSI the *refresh floor* is the largest period meeting target, $P^*(f_D) = \max\{P : \text{FC} \leq \beta\}$.

Policies. The *naive* certificate selects the highest MCS clearing the eMBB target from a (fresh or held) CQI report. The *target-aware* URLLC policy selects a conservative MCS (a fixed SINR backoff) and uses $K=3$ frequency/spatial diversity branches (a block errs only if all branches err); a backoff alone cannot reach a 10^{-3} tail on a single branch, so the honest URLLC certificate combines both. The *OLLA* policy (§IV) adds an SINR offset driven by HARQ ACK/NACK to a held report.

Discipline. Each experiment is pre-registered with numerical bars and manipulation checks fixed before the graded run; the sealed record is content-hashed and graded on seeds disjoint from those used to design it, and negative or failing outcomes are kept [5]. The numbers below are the graded, sealed values (Table II).

III. CONSUMER-RELATIVITY IN THE RELIABILITY TARGET

A base station serving mixed traffic derives one CSI estimate per user and selects an MCS. Consider the same estimate consumed under two targets: eMBB ($\text{BLER} \leq 10^{-1}$) and URLLC ($\leq 10^{-3}$). We drive a real NR link at high mobility, apply the naive CSI-derived MCS, and measure the achieved BLER against each target; we then compare a *target-aware* policy that sizes its margin to the consumer’s target.

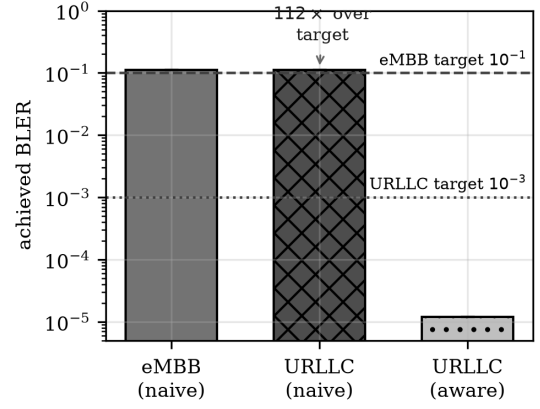


Fig. 1. Consumer-relativity of the CSI certificate in the reliability target (real NR, Sionna LDPC; three sealed seeds). The identical CSI-derived MCS meets the eMBB target (10^{-1}) but false-clears the URLLC target (10^{-3}) by $112\times$; a target-aware policy meets URLLC at the cost of ~ 2 dB margin.

Figure 1 reports the sealed result (three graded seeds). The naive certificate achieves $\text{BLER} \approx 0.112$. Read by the eMBB consumer this is at target—the certificate is sound. Read by the URLLC consumer the *same* 0.112 is a false-clear by $112\times$ (against 10^{-3}): every such block is a reliability-SLA violation, invisible to a scheduler that assumed one CSI report serves all traffic. The target-aware policy—the same CSI, a conservative MCS backoff combined with the $K=3$ diversity branches introduced in Sec. II (a backoff alone cannot reach the 10^{-3} tail on one branch)—achieves $\text{BLER} \approx 1.2 \times 10^{-5}$, comfortably within target, at the cost of the backoff’s spectral efficiency and the diversity’s resource overhead. No single certificate is correct for both consumers: the eMBB-adequate one fails URLLC, and the URLLC-safe one over-provisions eMBB. The false-clear rate is the number that prices the difference.

IV. THE REFRESH HORIZON

The second axis is time. A CSI report is fresh when issued and ages as the channel decorrelates; a certificate that was sound at report time false-clears once the channel has moved. At $f_D = 200$ Hz with a 20-TTI report period, the naive (held-CSI) certificate reaches $\text{BLER} \approx 0.36$ against a 0.1 target—a false-clear on roughly a quarter of transmissions—while *OLLA* (3), the deployed witnessed correction that adjusts an offset from HARQ feedback, holds it at ≈ 0.10 (Fig. 2(a) shows the aging across mobility).

The operational question is how often to refresh. Sweeping Doppler and report period, we define the *refresh floor* as the largest report period keeping BLER at target, and find it proportional to the Clarke coherence time $T_{\text{coh}} = 0.423/f_D$:

$$\text{refresh floor} \approx 0.177 T_{\text{coh}}, \quad R^2 = 0.915, \quad (4)$$

across an order of magnitude of mobility (Fig. 2(b)). At high Doppler the floor saturates at the minimum report period (one TTI): the certificate cannot be refreshed faster than the frame

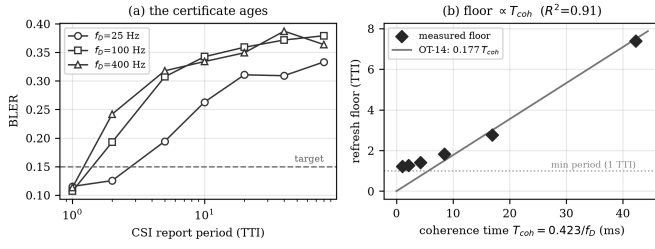


Fig. 2. The CSI certificate’s refresh horizon (real NR, Sionna LDPC; sealed sweep). (a) The certificate ages: BLER grows with report period, faster at higher Doppler. (b) The refresh floor is proportional to the Clarke coherence time ($0.177 T_{coh}$, $R^2 = 0.915$); at high Doppler it saturates at one TTI.

TABLE II
SEALED PER-SEED RESULTS (ACHIEVED BLER)

(a) Reliability target (§III), fresh CSI				
seed	eMBB	URLLC-naive	URLLC-aware	
20260824	0.1119	0.1119	1.7×10^{-5}	
20260825	0.1103	0.1103	1.9×10^{-5}	
20260826	0.1125	0.1125	$< 10^{-5}$	
(b) Refresh horizon (§IV), $f_D = 200$ Hz, 20-TTI				
seed	naive	OLLA	fresh	CQI err
20260823	0.366	0.101	0.116	5.61 dB
20260824	0.340	0.102	0.115	5.86 dB
20260825	0.373	0.101	0.113	5.76 dB

allows, so above a mobility threshold no reporting cadence suffices and a witnessed (HARQ-driven) correction becomes mandatory rather than optional. Equation (4) gives a concrete provisioning rule: re-report before the channel decorrelates past $\approx 0.18 T_{coh}$, not on a fixed calendar.

V. RELATED WORK

Link adaptation, CSI aging, and outer-loop correction are mature. OLLA [1] adjusts an SINR offset from HARQ ACK/NACK to hold a BLER target under imperfect or stale CSI; it is precisely the witnessed correction our refresh-horizon result quantifies. Short-packet and URLLC analysis [3] establishes the reliability-target regime in which the eMBB-to-URLLC gap of §III lives. Learned CSI feedback trained on reconstruction error [4] is an active direction; we note only that the reconstruction-versus-consumer objective gap it might imply did not appear at practical operating points in our controlled tests, and we make no claim on it here. Our contribution is orthogonal to all three: not a better estimator, scheduler, or offset rule, but a *measurement*—the false-clear rate as a first-class, consumer-relative KPI—and the two relativities and the refresh law it exposes.

VI. DISCUSSION AND CONCLUSION

The unit of correctness in link adaptation is not the CSI report but the pair (report, consumer). Sizing margin and refresh cadence for the consumer’s reliability target and coherence time, rather than for a nominal channel, is what keeps a certificate honest. The false-clear rate makes the cost of ignoring the consumer measurable: $112\times$ for a mismatched target, a coherence-scaled horizon for a stale report. The

Sionna link-level chain is the evidence rung; an over-the-air or srsRAN testbed with real HARQ traces is the external-validity graduation. The methodology is not radio-specific—the same consumer-relative false-clear appears in optical quality-of-transmission and database replication [5]—but the results here stand on the 5G NR CSI certificate alone.

REFERENCES

- [1] A. Sampath, P. Sarath Kumar, and J. M. Holtzman, “On setting reverse link target SIR in a CDMA system,” in *Proc. IEEE 47th Veh. Technol. Conf. (VTC)*, vol. 2, 1997, pp. 929–933.
- [2] J. Hoydis *et al.*, “Sionna: An open-source library for next-generation physical layer research,” arXiv:2203.11854, 2022.
- [3] G. Durisi, T. Koch, and P. Popovski, “Toward massive, ultrareliable, and low-latency wireless communication with short packets,” *Proc. IEEE*, vol. 104, no. 9, pp. 1711–1726, 2016.
- [4] C.-K. Wen, W.-T. Shih, and S. Jin, “Deep learning for massive MIMO CSI feedback,” *IEEE Wireless Commun. Lett.*, vol. 7, no. 5, pp. 748–751, 2018.
- [5] A. H. Bond, “Observation Theory: consumer-relative witnessed certificates,” sealed pre-registrations and code, <https://github.com/ahb-sjsu/observation-theory-campaigns>, 2026.